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# THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

## *Screening Test - Bhaskara Contest*

*(NMTC-Maths Olympiad at* ***JUNIOR LEVEL*** *IX & X Standards)*

*Saturday, 26th August 2006.*

*Note:*

1) *Fill in the answer sheet with your Name, Class, and Section etc. in the specified places.*
2) *Only one of the choices A,B,C,D is correct for each question. Write the alphabet of your choice in the answer sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).*
3) *For each correct response you get 1 mark;* ***for each incorrect response you lose ¼ mark.***
4) *Diagrams are only visual aids; they are not drawn to scale.*
5) *You are free to do rough work on separate sheets.*
6) *Duration of the test: 2 p.m. to 4. p.m.- 2 hours*

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1. In an n sided regular polygon the radius of the circum-circle is equal in length to the shortest diagonal. Then n is
   **(A)** 6 **(B)** 8 **(C)** 12 **(D) Such a polygon does not exist.**

2. All the six faces of a cube are extended in all directions. The number of regions into which the whole space is divided by these 6 planes is
   **(A) 9** **(B) 16** **(C) 24** **(D) 27**

3. Let $A = a + a^2 + a^3 + \ldots + a^{2006}$, and $a = -1$, then $\dfrac{A^3 - a^2}{A^2 - a}$ is a
   **(A) positive integer** **(B) negative integer**
   **(C) negative fraction** **(D) positive fraction**

4. How many possible values can one get by multiplying 2 different numbers from the set $\{4, 8, 9, 16, 27, 32, 64, 81, 243\}$?
   **(A) 12** **(B) 24** **(C) 36** **(D) 48**

5. Define $a * b = 2a + 2b - ab$. For example, $4 * 3 = 2x4 + 2x3 - 4x3 = 2$. If $3 * x = 2 * x$, then x is
   **(A)** 2 **(B)** 1 **(C)** 0 **(D) No such x exists**

6. If ABCD = (ABC)x(BD), where A,B,C and D are digits not necessarily distinct, then the value of C is
   **(A) 0** **(B) 1** **(C) any digit** **(D) any nonzero digit**

7. If. $a * b = a + b + 1$ and if $a * 5 = b * 4 = 11$, then $a * b$ is
   **(A) 10** **(B) 11** **(C) 12** **(D) 13**

8. a, b and c are three natural numbers. Exactly two of them are odd. Then

**(A) $a + b + c + ab + bc + ca$ is odd** **(B) $ab + bc + ca$ is even**

**(C) $a^3bc$ is odd** **(D) $a^2 + bc$ is even**

9. Given $x + 3y = 100$ where x and y are positive integers. The number of pairs satisfying the above equation is

**(A) 100** **(B) 97** **(C) 34** **(D) 33**

10. A three digit number with digits A, B, C in that order is divisible by 9. A is an odd digit and C is an even digit. B and C are non zero. The number of such three digit numbers is

**(A) 20** **(B) 16** **(C) 8** **(D) 4**

11. The number of prime numbers less than 1 million whose digital sum is 2 is,

**(A) 5** **(B) 4** **(C) 3** **(D) none of these**

12. The value of $\dfrac{36^{24}}{24^{36}}$ on simplification is

**(A)** $\left(\dfrac{3}{2}\right)^{\frac{1}{2}}$ **(B)** $\dfrac{3}{2^{12}}$ **(C)** $\dfrac{3^2}{2^3}$ **(D)** $\left(\dfrac{3}{32}\right)^{12}$

13. In Eldorado, the base in the number system is b, unlike our decimal system where 235 means $2.10^2 + 3.10 + 5$. The place values in Eldorado are powers of b and the currency is in dinars. Ram gave a 1000 dinar bill for a bicycle costing 440 dinars and receives 340 dinars as change. The base b is

**(A) 12** **(B) 8** **(C) 7** **(D) none of these**

14. A certain number leaves a remainder 4 when divided by 6. The remainder when the number is divided by 9 is

**(A) 1 or 4** **(B) 4 or 7** **(C) 1 or 7** **(D) 1, 4 or 7**

15. Given the alphametic where each letter represents a different digit, the number of distinct solutions is

$$\begin{array}{r} ABCD \\ -\ BCDA \\ \hline 1512 \end{array}$$

**(A) 0** **(B) 1**

**(C) 6** **(D) 7**

16. In a $\Delta ABC$, $\angle A = 30^o$, AC = 10 units and BC = 4 units. Which of the following statements is correct?

**(A) $\angle B$ is acute** **(B) $\angle B$ is obtuse**

**(C) $\angle B$ is $90^o$** **(D) such a triangle does not exist**

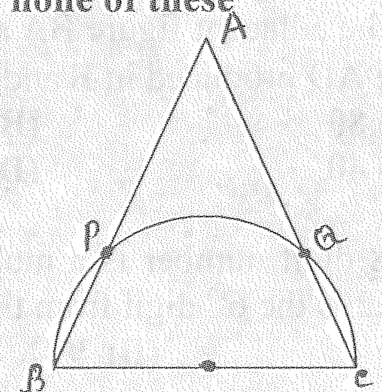
17. The number of solutions (x, y) where x and y are integers, satisfying $2x^2 + 3y^2 + 2x + 3y = 10$ is

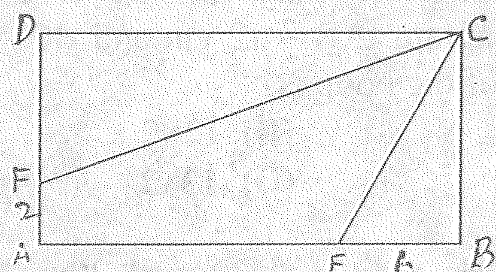
**(A) 0** **(B) 2** **(C) 4** **(D) none of these**

18. BC is the diameter of a semicircle. The sides AB and AC of a triangle ABC meet the semicircle in P and Q respectively. PQ subtends $140^o$ at the centre of the semi- circle. Then $\angle A$ is

**(A) $10^o$** **(B) $20^o$**

**(C) $30^o$** **(D) $40^o$**




19. 


In rectangle ABCD, $\angle C$ is trisected by CE and CF where E lies on AB and F on AD. If BE = 6 cm and AF = 2 cm, which of the following integers is nearest to the area of the rectangle ABCD in sq.cm.?

**(A) 130** **(B) 150**

**(C) 170** **(D) 190**

20. The first two terms of a sequence are 0 and 1. The nth term $T_n = 2T_{n-1} - T_{n-2}$, $n \geq 3$. For example the third term $T_3 = 2T_2 - T_1 = 2 - 0 = 2$. The sum of the first 2006 terms of this sequence is

(A) $\frac{2006x2007}{2}$ (B) $\frac{2005x2006}{2}$ **(C) 2006** **(D) 2005**

21. Eighteen balls are placed in 4 boxes so that no box is empty. Then

**(A) At least one box has at least 5 balls** **(B) At least one box has exactly 5 balls**

**(C) Exactly one box has at least 5 balls** **(D) Exactly one box has exactly 5 balls**

22. The least number of numbers to be deleted from the set {1, 2, 3,…,13, 14, 15} so that the product of the remaining numbers is a perfect square is

**(A) 4** **(B) 3** **(C) 2** **(D) 1**

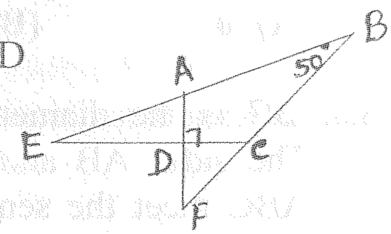
23. If $(43)_x$ in base x number system is equal to $(34)_y$ in base-y number system then the values for x and y are respectively

**(A)** 3, 4 **(B)** 4, 3 **(C)** 7, 9 **(D)** 9, 7

24. Consider the following sequence: $a_1 = a_2 = 1$, $a_i = 1 + \text{minimum}\{a_{i-1}, a_{i-2}\}$ for $i > 2$. Then $a_{2006} =$

**(A)** 1003 **(B)** 1002 **(C)** 1001 **(D) none of these.**

25. In the adjacent figure BA and BC are produced to meet CD and AD produced in E and F. Then $\angle AED + \angle CFD$ is

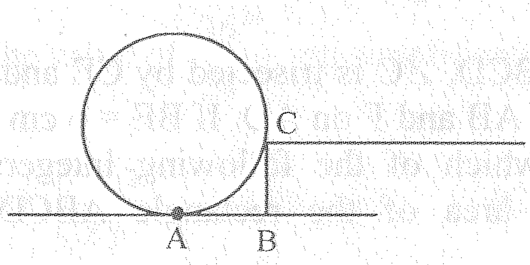



**(A)** $80^\circ$ **(B)** $50^\circ$

**(C)** $40^\circ$ **(D)** $160^\circ$

26. A 8 digit number is a multiple of 73 and 137. If the second digit from left is 7, what is the $6^{th}$ digit from the left of the number?

**(A)** 7 **(B)** 9 **(C)** 5 **(D)** 3

27. 


A hoop is resting vertically at stair case as shown in the diagram. AB = 12 cm and BC = 8 cm. The radius of the hoop is

**(A)** 13 **(B)** $12\sqrt{2}$

**(C)** 14 **(D)** $13\sqrt{2}$

28. Harsha looks at a calendar for the year 20xy. He notices that April 20xy has exactly 4 Mondays and 4 Fridays. Then 28 April 20xy would fall on a

**(A) Sunday only** **(B) Friday, Monday or Tuesday only**

**(C) Monday only** **(D) Monday, Sunday or Tuesday only**

29. In a triangle ABC, P and Q are midpoints of AB and AC respectively. R is a point on BC such that BR = 2RC. Let S be the intersection of PQ and AR. If the area of triangle AQS is 1 sq. unit, then the area of trapezium PSRB is

**(A)** 6 **(B)** 8 **(C)** 9 **(D)** 12

30. A solid cube is chopped off at each of its 8 corners to create an equilateral triangle with 3 new corners. The 24 corners are now joined to each other by diagonals. How many of these diagonals completely lie in the interior of the cube?

**(A)** 36 **(B)** 96 **(C)** 108 **(D)** 120